

Reference Architecture RA-01 — Predictive Maintenance at 1,000 Sensor Channels, 100% On-Premises

Audience: VP of IT/OT and the architecture review board · **Status:** For approval **Companion document:** RA-02 — Hybrid with AWS-hosted services (identical section numbering for side-by-side comparison) **Version basis:** SUSE Edge 3.6.1 release matrix, verified against documentation.suse.com on 2026-07-23 · **Re-verify by:** 2026-10-23

1. Executive summary

Decision requested: approve a two-phase deployment of an on-premises predictive-maintenance platform for the site's ~1,000 sensor channels — Phase 1 pilot on one production line (8 weeks, exit criteria in §13), Phase 2 site rollout contingent on pilot exit — with budget per the cost model in §12 and the staffing plan in §10.6.

Situation. The site's process tools generate continuous sensor telemetry that today is used reactively: excursions are found after scrap or downtime, and condition data leaves value on the table. Unplanned downtime is the single largest recoverable manufacturing loss — across industry, Siemens' *True Cost of Downtime 2024* estimates it at ~\$1.4T annually for the world's 500 largest companies (~11% of revenues), and in ITIC's 2024 survey more than 90% of mid-size and large enterprises put one hour of downtime above \$300,000. This site's own figure is an input this document requests (§12).

Complication. The telemetry that enables prediction is also among the site's most sensitive intellectual property: process traces and recipes reveal how we make what we make. Any solution that ships raw telemetry off-site is a non-starter — which rules out the default cloud-analytics pattern, and is why this architecture is designed *from* the sovereignty constraint rather than patching it on.

Answer. Deploy scoring at the edge: a hardened, highly available Kubernetes platform on the plant floor runs statistical process control (Hotelling T²/EWMA) models against every channel in real time, forecasts remaining useful life, attributes excursions to specific sensors, and keeps **every byte of raw telemetry inside the site boundary — zero WAN egress in this variant**. The platform is built from a certified, co-versioned open-source stack (SUSE Edge 3.6.1: SUSE Linux Micro, RKE2/K3s, Rancher Prime, Longhorn, NeuVector; SUSE Industrial Edge gateways; on-premises platform install) with a documented exit path at every layer (§14). The detection pattern is not speculative: a runnable reference kit implements this exact pipeline end to end and is verified in CI, including a golden-parity test of the scoring model and a live egress-block exercise (§15.1 — with explicit scope statements about what those proofs do and do not cover).

Business case (sourced, to be localized with site actuals). McKinsey's operations practice reports predictive maintenance typically reduces machine downtime 30–50% and increases machine life 20–40%; Deloitte's position paper reports on average +25% productivity, –70% breakdowns, –25% maintenance cost. Applied to this site's downtime cost (\$/hour — required input, §12), even the conservative end funds the platform severalfold. No site-specific ROI is asserted in this document; §12 provides the formula and a clearly-labeled illustrative example.

Why this design makes sense (and would with different logos): one declarative platform for all plant-floor workloads instead of per-tool appliances; immutable transactional OS with rollback for change-safe maintenance windows; zero-trust runtime security with an auditable egress contract; fleet-consistent operations that scale from this site to others; long support horizons (base OS general support to 2029-11-30) with single-vendor accountability across the stack.

2. Scope, assumptions, and claim labeling

2.1 Claim labeling used throughout

Label	Meaning
[Lab-verified]	Demonstrated in the reference kit in this repository; scope stated at each citation (§15.1)
[Sourced]	Vendor documentation or named study; version and verification date pinned
[Assumption A#]	Design assumption, numbered in §2.2; every derived figure cites its assumption
[FILL]	Site-specific input this document requests before final costing

2.2 Assumption log

#	Assumption	Sensitivity
A1	"1,000 sensor endpoints" = 1,000 discrete sensor channels across ~40–60 process tools, scalar telemetry at 1 Hz plus windowed derived features. High-rate waveform channels (e.g., kHz vibration) are out of scope for the base design; if present, feature extraction moves adjacent to the tool and the ingest tier is re-sized.	Critical — every throughput and sizing number derives from this
A2	Message shape ≈ 200 bytes/sample ⇒ ~1,000 msg/s site-wide, ~86.4M msgs/day, ~0.2 MB/s sustained ingest.	Medium
A3	Alarm/query rate for the optional AI-explanation tier ≤ 50 interactions/hour site-wide (it scales with excursions and engineer queries, not with sensor count).	Medium (affects only optional Tier 3)
A4	Site unplanned-downtime cost [FILL: \$/hour from finance]. Industry reference points in §12 only.	Critical to ROI, not to architecture
A5	Brownfield site: existing SCADA/historian/FDC systems remain; this platform ingests via OPC UA/Modbus alongside them (§4.4).	High
A6	One site, fleet-ready: management plane sized for multi-site expansion, but all sizing and cost in this document are single-site.	Medium

2.3 In scope / out of scope

In scope: sensor aggregation, edge scoring, alerting/visualization, optional on-prem AI explanation, platform security, HA, Day-2 operations. **Out of scope:** closed-loop control (this platform advises maintenance; it does not command tools), MES/ERP integration beyond alert export, replacing incumbent FDC/SCADA (§4.4).

3. Requirements

3.1 Functional (condensed)

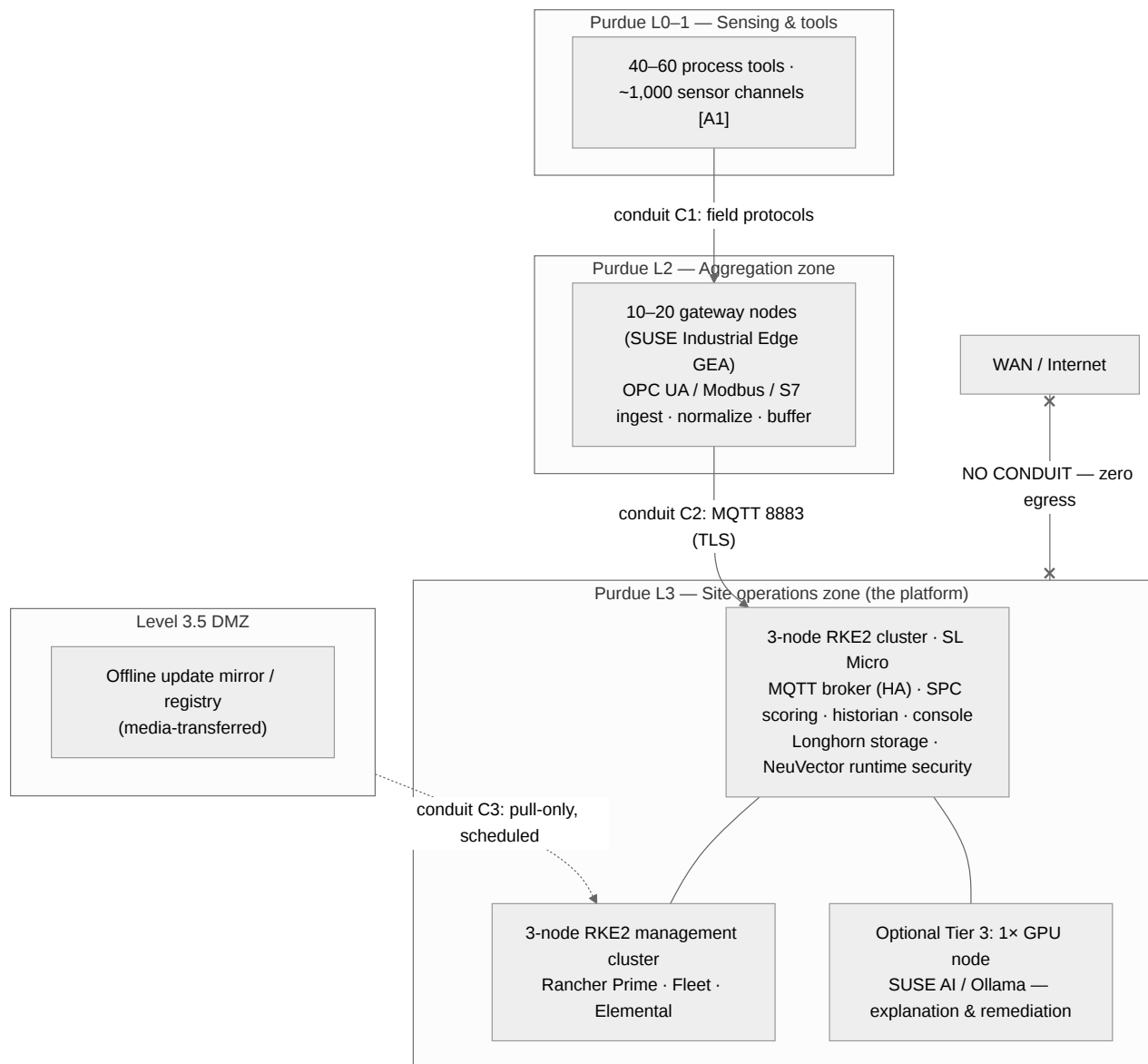
Continuous per-channel scoring with per-tool learned baselines; excursion detection with sensor attribution; remaining-useful-life forecasting; operator console with alarm management; fault-injection test harness for commissioning; on-prem AI-generated remediation guidance (optional tier).

3.2 Non-functional requirements

NFR	Target	Verification method
Data sovereignty	Zero WAN egress from the OT zone. Raw telemetry, derived scores, and configuration never leave the site	Firewall ruleset review + §7 defense-in-depth + quarterly egress audit (§10.3); mechanism demonstrated by the kit's <code>sovereignty-verify</code> [Lab-verified, single-namespace scope]
Availability behavior	No single point of failure in ingest, scoring, or storage; behavior under each failure mode per §8 table (no unsourced "nines" are claimed)	Failure-mode drills at pilot exit (§13)
Recovery targets	Node loss: automatic, ≤ 5 min service restoration target. Quorum loss: ≤ 4 h restore-from-snapshot target. Site power loss: cold-start runbook ≤ 1 h target	Timed drills, §13
Data retention	90 days full-resolution on-site [Assumption A2 sizing]; 3 years derived health history	Storage audit
Scoring latency	Sensor sample → scored verdict ≤ 5 s end-to-end	Pilot measurement
Security	IEC 62443-3-3 SL 2 target (SL-T) for the monitoring zone; zones/conduits per §7	Control-mapping table §7.5 + third-party assessment [FILL: schedule]
Patching	Monthly OS/platform patch window with A/B rollback; emergency CVE path ≤ 72 h	§10.1 process; drill at pilot

4. Architecture overview

4.1 Zone model (Purdue / IEC 62443)



4.2 Tiers

Tier	Function	Components (product in parentheses)
Sensing	Existing tool sensors and PLCs	Site equipment (unchanged)
Aggregation	Protocol ingest, normalization, offline buffering, the single governed egress point design pattern	Gateway Edge Agents (SUSE Industrial Edge / Losant GEA — 512 MB/8 GB min per agent, 65,000-message offline buffer [Sourced: docs.losant.com])
Platform	HA Kubernetes, storage, runtime security	3-node RKE2 1.35.4 on SL Micro 6.2; Longhorn 1.11.2 (3-replica, dedicated SSD + storage network); NeuVector 5.5.2 (3 controllers, enforcer per node) [Sourced: SUSE Edge 3.6.1]

Tier	Function	Components (product in parentheses)
Scoring	Per-channel SPC (T ² /EWMA), RUL, attribution	Containerized scoring service — the reference kit's model [Lab-verified: golden parity; fidelity, not throughput]
Data & console	HA MQTT, time-series historian, operator console, alarm management	Clustered MQTT broker (see §5 alternatives), on-prem historian, on-prem platform console (Losant on-premises install — vendor documents on-prem/private-cloud deployment [Sourced])
Management	Provisioning, GitOps, lifecycle	Rancher Prime 2.14.2 + Fleet on a separate 3-node RKE2 management cluster; Edge Image Builder 1.3.3.1 + Elemental 1.9.0 [Sourced]
Optional Tier 3	On-prem AI explanation/remediation	1× GPU node (SUSE AI floor: 8–16 cores / 64 GB / 100 GB SSD + NVIDIA GPU [Sourced]); sized by alarm rate [A3], not sensor count. The SPC tier operates fully without it

4.3 Design properties worth approving

- **Agents dial out, nothing dials in:** downstream clusters initiate all connections to the management plane [Sourced: Rancher docs]; combined with the zero-egress rule this means no inbound firewall holes to the OT zone.
- **Declarative recovery:** every workload and cluster is Git-defined; a destroyed node is re-imaged (Elemental) and converges automatically.
- **Immutable OS with rollback:** SL Micro A/B transactional updates make patch windows reversible by design.

4.4 Brownfield coexistence (mandatory posture)

This platform **augments** the incumbent FDC/SCADA/historian estate; it does not replace it. Ingest is read-only from existing OPC UA servers/historians where available, direct from PLC protocols where not. Alerts export to the existing CMMS/alarm workflow. Rip-and-replace is explicitly rejected (§14.1). [Assumption A5]

5. Component catalog, alternatives, and exit paths

Function	Selected	Rationale (function first)	Primary alternative	Exit path
Edge OS	Immutable transactional Linux (SL Micro 6.2)	A/B rollback, SELinux enforcing, general support to 2029-11-30 [Sourced: suse.com/lifecycle]	Flatcar / RHEL for Edge	Standard OCI workloads move unchanged
Kubernetes	RKE2 1.35.4 (etcd, CIS/FIPS-capable)	Hardened profile + 3-node quorum; K3s reserved for the pilot footprint	Vanilla k8s / OpenShift	CNCF-conformant — manifests portable

Function	Selected	Rationale (function first)	Primary alternative	Exit path
GitOps/mgmt	Rancher Prime + Fleet	Fleet-consistent config; dial-out agents	Argo CD + upstream Rancher	Continue on OSS unsupported, or Argo
Storage	Longhorn 1.11.2, 3-replica	Replicated block on commodity SSD; snapshot/backup native	Rook/Ceph	CSI standard; PV migration
Runtime security	NeuVector 5.5.2	L7 zero-trust, Protect mode, admission control; one layer of \$7's stack	Cilium+Falco assembly	Policies re-expressible; OSS NeuVector exists
MQTT	Design choice at detail phase: clustered broker (EMQX/HiveMQ) for single logical HA endpoint, or per-cell Mosquitto with gateway failover — Mosquitto itself does not cluster [Sourced]	1,000 msg/s is trivial for either [A2]; the choice is an HA-topology decision, not throughput	—	MQTT 3.1.1/5 standard protocol
Gateways	SUSE Industrial Edge GEA	OPC UA/Modbus/S7/EtherNet-IP ingest, offline buffer, workflow engine [Sourced]	Telegraf/Node-RED assembly	MQTT contract preserved
Platform console	Losant on-premises install	Vendor-documented on-prem deployment option [Sourced]; single pane for device/dashboard/workflow	Grafana + custom	Historian data is open time-series
Scoring	Reference-kit SPC service	Open Python/stdlib model, golden-parity CI [Lab-verified]	Commercial FDC add-on	The model is ours; no lock-in

6. Data governance & sovereignty

6.1 Data classification

Class	Examples	Residency
C1 — Trade secret	Raw sensor traces, recipes, tool parameters, lot ↔ tool mapping	Never leaves the OT zone. Not the DMZ, not IT, not anywhere
C2 — Operationally sensitive	Derived health scores, RUL, attribution, alarm history	Site operations zone; visible to site engineering
C3 — Administrative	Platform configs (secrets externalized), software inventories, logs w/o payloads	Site; DMZ mirror for update metadata only

6.2 Egress matrix (the auditable artifact)

Flow	Content	Direction	Allowed?
Tools → Gateways	C1 raw	L0/1 → L2	Yes (conduit C1)
Gateways → Platform	C1 raw (normalized)	L2 → L3	Yes (conduit C2, TLS)
Platform → anywhere beyond L3	any	L3 → DMZ/IT/WAN	No. Zero egress.
DMZ mirror → Management	Update artifacts (signed, media-loaded)	DMZ → L3	Pull-only, scheduled (conduit C3)

Updates enter by removable media or a one-way staged mirror; nothing exits. The gateway tier's governed-egress design (only derived fields *could* ever be published) remains in force as defense-in-depth even though this variant permits no egress at all — the mechanism is demonstrated by the kit [Lab-verified: air-gapped counter behavior; single-node scope].

7. Security architecture

7.1 Defense-in-depth stack (no single product carries the claim)

1. **Physical/VLAN segmentation** — OT zones per Purdue; managed switches, port security.
2. **Site firewall** — default-deny both directions at every conduit; the L3 → WAN conduit **does not exist**.
3. **Kubernetes NetworkPolicy** — default-deny egress in workload namespaces [Lab-verified mechanism: `make sovereignty-verify`, blocked egress with positive control].
4. **Runtime L7 enforcement** — NeuVector Protect mode: learned allowlists, process/file profiles, admission control. *Protect-mode behavior at production scale is validated in Phase 1 (§13) — the lab could not exercise it in nested containers and this document does not claim it as lab-proven.*
5. **Governed egress application design** — the gateway is the only component that could publish beyond the broker, and its schema is derived-only.

7.2 Identity & access

Site IdP (AD/LDAP) → Rancher RBAC for platform roles; NeuVector and console SSO against the same IdP; per-namespace service accounts, no shared admin; break-glass procedure with sealed credentials [FILL: site process].

7.3 Secrets, certificates, supply chain

Secrets in-cluster (sealed at rest) with rotation calendar (§10.3); internal PKI for MQTT TLS and platform certs; images from the site registry only — signed, provenance-checked (base stack ships with SLSA-4 build provenance, FIPS 140-3 modules available, Common Criteria EAL4+ base OS [Sourced: suse.com/security/certifications/]); admission control blocks unsigned images.

7.4 Patching & CVE response

Monthly window (§10.1); emergency path ≤ 72 h target using A/B rollback as the safety net. Vulnerability data reaches the air-gapped scanner via the DMZ mirror on the same cadence. (No numeric vendor CVE-fix SLA is published; none is claimed.)

7.5 IEC 62443 SL-2 control mapping (excerpt; full table at detail design)

62443-3-3 requirement (SR)	Control here
SR 1.x Identification & authentication	IdP-backed RBAC; unique service identities; TPM-rooted node identity (Elemental)
SR 3.x System integrity	Immutable OS, signed images, admission control, NeuVector process profiles
SR 5.x Restricted data flow	Zones/conduits §4.1; firewall + NetworkPolicy + L7 enforcement §7.1
SR 7.x Resource availability	§8 failure modes; UPS; capacity monitoring §10.2

Target is **SL-T 2** for the monitoring zone; this is a design target with mapped controls, **not** a certification claim. SEMI E187/E188 alignment noted for fab equipment cybersecurity context.

8. Availability & failure modes

No availability percentage is asserted (none would be sourced). Approval should rest on behavior:

Failure	Behavior	Recovery	Target
1 of 3 platform nodes lost	Workloads reschedule; Longhorn rebuilds replicas; ingest unaffected (gateway buffers absorb the reschedule gap)	Automatic; replace node via Elemental re-image	≤ 5 min service restoration; node replacement ≤ 1 day
2 of 3 nodes lost	etcd quorum lost: API/control plane down, no rescheduling or config changes; already-running pods on the survivor continue; some Longhorn volumes unavailable	Restore quorum from etcd snapshot per runbook; or rebuild + restore	≤ 4 h [target]
Single disk loss	Longhorn replica rebuild from healthy copies	Automatic	Rebuild window per volume size
Gateway node loss	That cell's channels buffer at the tool-side or drop per cell design; other cells unaffected	Spare gateway; Elemental re-image	≤ 2 h [target]
Broker partition	Clustered broker continues on quorum side; gateways buffer up to 65,000 msgs each [Sourced] and flush on heal — the kit demonstrates the buffer/flush mechanism [Lab-verified, single-gateway scope]	Automatic	—
Management cluster lost	Production scoring unaffected (management is control-plane only; agents reconnect when restored)	Restore mgmt cluster from backup	≤ 1 day [target]

Failure	Behavior	Recovery	Target
Site power event	UPS carries platform through generator start or graceful shutdown; cold-start runbook defines order (storage → broker → scoring → console)	Runbook	≤ 1 h from power-stable [target]

RPO: raw telemetry ≤ 60 s (gateway buffer + broker persistence); derived history ≤ 15 min (historian snapshot cadence). All targets are design targets to be demonstrated in Phase-1 drills (§13).

9. Sizing & bill of materials (derived from A1/A2)

Role	Qty	Spec (each)	Basis
Platform nodes	3	16 cores / 64 GB / 2× SSD (OS + 1 TB Longhorn) / 2× 10 GbE	Workloads + Longhorn best practice (4c/4GB for storage engine, SSD, 10 Gbps storage net) [Sourced: longhorn.io]
Management nodes	3	4 cores / 16 GB / SSD	Rancher "small" tier supports ≤150 downstream clusters — generous headroom [Sourced: Rancher 2.14 docs]
Gateway nodes	10–20	Dual-core / 2–4 GB (GEA min 512 MB/8 GB disk; 1 GB/dual-core recommended) [Sourced]	~50–100 channels per gateway, one per line/cell [Assumption — no vendor per-GEA cap is published]
Optional GPU node	1	16 cores / 64 GB / 100 GB SSD / 1× NVIDIA GPU	SUSE AI floor [Sourced]; sized by A3
Network	—	Dedicated storage VLAN 10 GbE; OT VLANs per zone	\$4.1
UPS	site-standard	Bridge to generator + graceful shutdown	\$8

Throughput sanity [A1/A2]: ~1,000 msg/s ≈ 0.2 MB/s sustained — an order of magnitude below single-broker capability; sizing is driven by HA and storage retention, not message rate. 90-day full-resolution retention ≈ 1.6 TB raw (0.2 MB/s × 86,400 × 90) before compression — within the 3×1 TB replicated pool with headroom [derived from A2].

10. Day-2 operations

10.1 Patching & lifecycle flow

Staged via Git: dev-ring cluster → pilot line → site. OS: SL Micro transactional update, reboot in maintenance window, automatic rollback on failed health checks. Platform: Rancher-orchestrated RKE2 upgrades (one node at a time, quorum preserved). Workloads: Fleet rollout with canary on the pilot line. Emergency CVE: same path, compressed to ≤ 72 h.

10.2 Observability

Platform metrics + alerting on-site (Prometheus-stack); scoring-tier KPIs (frames/s, verdict latency, model drift indicators); NeuVector security events into the site SIEM [FILL: SIEM integration detail].

10.3 Operations calendar

Cadence	Activities
Daily	Alarm review; backup job status; security-event triage
Weekly	Capacity trend review; patch-ring promotion check
Monthly	Patch window; egress-rule audit; restore-test one workload
Quarterly	Full DR drill (quorum-loss restore); access recertification; CVE-mirror refresh audit; model-performance review vs. maintenance outcomes
Annual	62443 control reassessment; lifecycle/EOL review (\$10.5); tabletop incident exercise

10.4 Backup & restore

etcd snapshots (both clusters) daily → on-site backup target (separate failure domain); Longhorn volume backups nightly; Git is the source of truth for all config (already off-node by design — on-site Git mirror); restore drills per calendar.

10.5 Lifecycle table (real dates — re-verify at approval)

Component	Version	Support horizon
SUSE Edge release stream	3.6.x	24-month release support [Sourced]
SL Micro	6.2	General support to 2029-11-30 [Sourced: suse.com/lifecycle]
RKE2/K3s, Rancher, Longhorn, NeuVector	per 3.6.1 matrix	Co-versioned within the Edge release; move with the stream

10.6 Staffing & RACI (roles, not headcount reductions — this platform feeds maintenance planning; it displaces no operator)

Activity	R	A	C	I
Platform admin & patching	Site platform engineer (new role, 1 FTE trained)	IT/OT VP	Vendor support	Site ops
Model & alarm tuning	Process engineering	Eng. manager	Platform eng.	Maintenance
Security operations	Site SecOps	CISO org	Platform eng.	IT/OT VP
Gateway/field maintenance	OT technicians	OT manager	Platform eng.	—
24×7 on-call	[FILL: site model — recommend platform eng. primary, vendor support escalation]	IT/OT VP	—	—

Vendor support entitlement (Rancher Prime subscription) provides the escalation path behind the on-call rotation; SLA terms per contract [FILL].

11. Risk register

#	Risk	L×I	Mitigation	Residual	Owner
R1	NeuVector Protect-mode behavior at scale unproven in our lab	M×M	Phase-1 gate validates on real 3-node RKE2 before any enforcement claim; NetworkPolicy + firewall enforce meanwhile	Low	Platform eng.
R2	K8s skills gap in OT team	H×M	EIB-baked images minimize hand-ops; Rancher UI; training in Phase 1; managed-service option priced as alternative	Medium	IT/OT VP
R3	Model false-alarm rate erodes trust	M×H	Pilot exit criterion caps false-alert rate; alarm tuning owned by process eng.; deterministic attribution shown with every alarm	Medium	Eng. manager
R4	On-prem platform console (Losant on-prem install) version/feature lag vs SaaS	M×L	Confirm on-prem feature matrix during procurement [FILL]; console is replaceable (\$5 exit path)	Low	Platform eng.
R5	Air-gap update logistics slip patch cadence	M×M	DMZ mirror + calendar discipline; quarterly audit	Medium	Site SecOps
R6	Incumbent-FDC overlap creates political/ownership friction	M×M	\$4.4 coexistence posture; joint pilot review with FDC owners	Medium	IT/OT VP
R7	Hardware supply lead times	M×M	Order pilot + spares at approval; commodity x86 avoids proprietary appliances	Low	Procurement

12. Cost model (formula, not fabrication)

This document does not assert a site ROI. It provides the structure and requests two inputs.

Annual benefit	= D × H_avoided	(D = downtime cost \$/hr [FILL A4]; H_avoided = hours avoided/yr – pilot-measured)
	+ M × r_m	(M = annual maintenance spend [FILL]; r_m = reduction rate – sourced range
25% [Deloitte])		
Annual cost	= HW amortization + subscriptions + 1 FTE platform eng. + power/space	

Illustrative only (assumption set A, not a promise): if D = \$500k/hr (a mid-range industry figure; semiconductor estimates run \$1M+/hr but **no authoritative primary source exists** — use site actuals)

and the pilot demonstrates even 10 avoided downtime hours/year, the benefit term is \$5M/yr against a platform cost measured in low hundreds of \$k/yr (hardware BOM \$9 at commodity pricing + subscriptions [FILL: quote] + 1 FTE). The decision economics survive an order-of-magnitude haircut on every assumption.

Sourced reference points for context: McKinsey 30–50% downtime reduction; Deloitte +25% productivity / –70% breakdowns / –25% maintenance cost; Siemens *True Cost of Downtime 2024*; ITIC 2024 (>\$300k/hr for >90% of enterprises surveyed).

13. Implementation roadmap

Phase	Duration	Content	Go/no-go exit criteria
0 — Approval & procurement	2–4 wks	This document; hardware order; on-prem console procurement incl. feature-matrix confirmation [R4]	Signed decision; BOM ordered
1 — Pilot (one line)	8 wks	3-node platform + 2 gateways + ~100 channels; brownfield ingest from existing historian; full security stack	Measured: detection lead time on seeded faults ≥ [FILL] hrs; false-alert rate ≤ [FILL]/day; gateway disconnect test (buffer/flush, zero loss); sovereignty verification witnessed by site security; node-loss drill meets §8 targets; NeuVector Protect-mode egress block demonstrated on real RKE2 [R1]
2 — Site rollout	12–16 wks	Remaining lines/gateways; alarm workflow integration; ops handover per RACI	All lines scoring; ops calendar running 1 full cycle; quorum-restore drill passed
3 — Optional Tier 3	+4 wks	GPU node + on-prem explanation tier	Engineer-rated usefulness ≥ agreed bar; A3 validated

Pilot deliberately reuses the verified reference kit as its starting point — commissioning is configuration, not invention [Lab-verified basis, §15.1].

14. Alternatives considered & exit strategy

14.1 Alternatives

Option	Why not selected
Cloud analytics (any hyperscaler)	Violates the sovereignty constraint for C1 data by design; see RA-02 for the bounded hybrid that doesn't
Incumbent FDC expansion	Complements rather than competes (§4.4); doesn't provide fleet-consistent platform, modern runtime security, or the open scoring model; quote invited as benchmark [FILL]

Option	Why not selected
Azure IoT Operations / AKS-EE pattern	Credible architecture; not selected because this design requires zero hyperscaler control-plane dependency in the OT loop and an air-gap-first update path. Recorded honestly, not strawmanned
Per-tool vendor appliances	Fragmented security/patching surface; no unified fleet ops; per-tool licensing scales badly to 1,000 channels
VM-only (no Kubernetes)	Loses declarative recovery, canary rollouts, and fleet consistency; concedes §4.3 properties that motivate the design

14.2 Lock-in / exit analysis (per layer)

Protocols are open (OPC UA, MQTT, CSI, CNCF K8s); the scoring model is site-owned open code; exit from commercial subscriptions = continue on the open-source upstreams unsupported (Rancher/Longhorn/NeuVector all have OSS lines) or substitute per §5 exit paths. Proprietary surface is limited to: supported builds/entitlements, NeuVector enterprise features, and the platform console's workflow definitions (exportable; historian data open).

15. Appendices

15.1 Evidence base (what is proven, and what it does — and does not — prove)

Artifact (this repository)	Proves	Does prove
Runnable kit (<code>make up</code> , fault inject/heal, end-to-end)	The full pattern operates: ingest → score → alarm → heal, single node	Scale, throughput, availability
Golden parity (590 recorded frames, every verdict field CI-compared)	Scoring-model fidelity across implementations	Any throughput/scale property
<code>make sovereignty-verify</code> (blocked egress + positive control)	The egress-enforcement mechanism , one namespace, lab	Site-wide enforcement at scale
Live management wiring (Rancher import + GitOps deploy, verified)	The Day-2 management mechanism	Multi-site fleet operations
Live demo console (golden-parity model in-browser)	Stakeholder-visible behavior of the exact scoring logic	Anything about production infrastructure

15.2 Glossary

SPC (statistical process control) · T^2 (Hotelling multivariate statistic) · EWMA (exponentially weighted moving average) · RUL (remaining useful life) · FDC (fault detection & classification) · GEA (Gateway Edge Agent) · Purdue model (ISA-95 network segmentation levels) · SL-T (IEC 62443 target security level) · A/B updates (dual-partition transactional OS upgrades).

15.3 References

SUSE Edge 3.6.1 release notes (documentation.suse.com, verified 2026-07-23) · Rancher v2.14 installation requirements · Longhorn 1.11 best practices · NeuVector production deployment docs ·

SUSE AI 1.0 requirements · docs.losant.com (GEA specs; on-premises install) · suse.com lifecycle & security certifications (SL Micro 6.2; EAL4+, FIPS 140-3, SLSA-4) · McKinsey Operations, *Manufacturing: Analytics unleashes productivity and profitability* · Deloitte Analytics Institute, *Predictive Maintenance* position paper (2017) · Siemens (Senseye), *The True Cost of Downtime 2024* · ITIC, *2024 Hourly Cost of Downtime Survey* · IEC 62443-3-3 · NIST SP 800-82r3 / NISTIR 8183r2 · SEMI E187/E188.